

Fake News Detection on **Twitter**

Using Natural Language Processing and Machine Learning to Classify Misinformation

The Misinformation Crisis

Twitter enables rapid information sharing, but it also accelerates the spread of fake news—creating confusion and eroding trust.

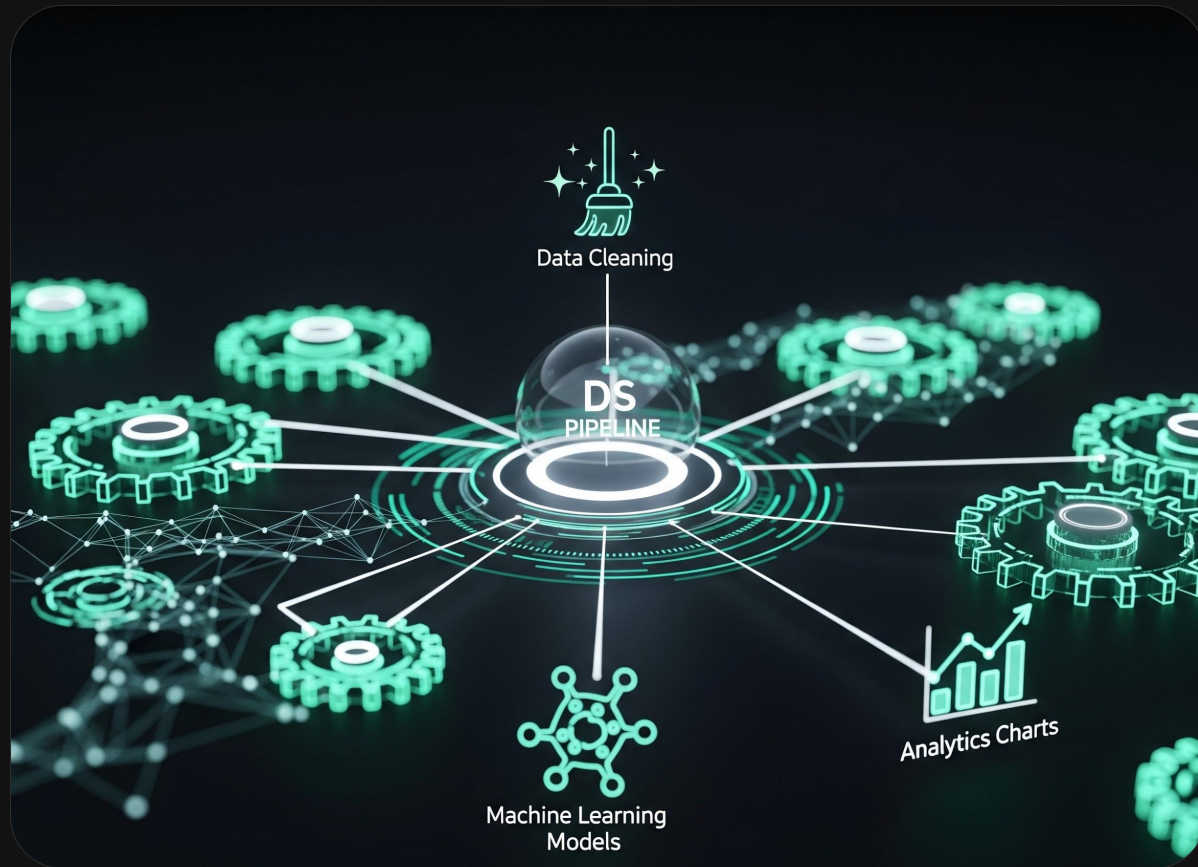
The Challenge: Millions of tweets daily contain unverified claims.

Manual fact-checking cannot keep pace.

The Goal: Build an automated ML classifier to label tweets as "Real" or "Fake" in real-time.



Data Science Workflow



Preprocessing

Cleaning raw textual noise.



Feature Extraction

Converting text to vectors.



Modeling

Training classifiers for prediction.



Evaluation

Measuring accuracy and reliability.

Stage 1: Data Preprocessing



Text Cleaning

Removing URLs, hashtags, punctuation, and special symbols that don't contribute to sentiment.



Normalization

Converting all text to lowercase and performing tokenization to break sentences into individual words.



Refining

Removing common stopwords and applying Stemming/Lemmatization to reach root word forms.

Stage 2: **Feature** Extraction

TF-IDF Vectorization

Reflects how important a word is to a document in a corpus. It balances high-frequency words vs unique discriminators.

CountVectorizer

A "Bag of Words" approach that creates numerical representations based on simple word frequency counts.

Word Embedding

Captures semantic relationships and context by mapping words to dense vectors in a high-dimensional space.

The **Model** Landscape



Logistic Regression

A simple baseline model that performs exceptionally well on high-dimensional sparse text data.



Naive Bayes

A fast, probabilistic classifier based on Bayes' theorem, ideal for multi-class text problems.



SVM

Support Vector Machines find the optimal hyperplane to separate complex patterns in linguistic features.

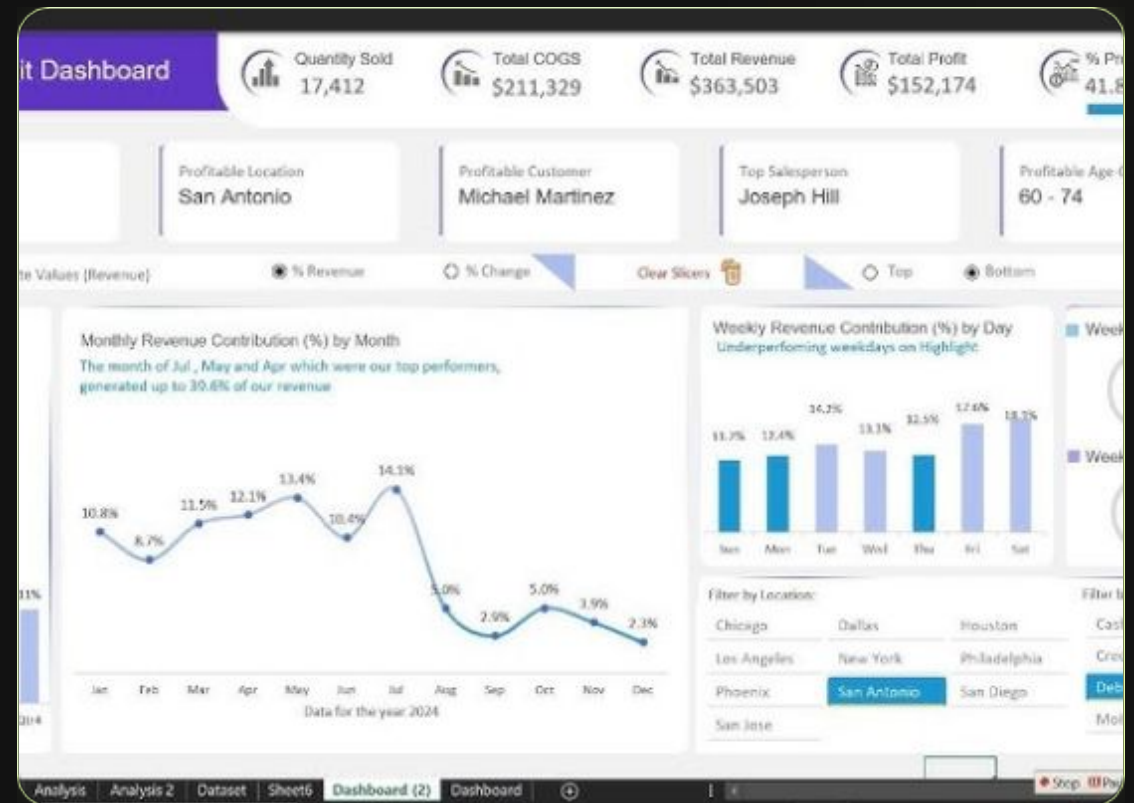
Model Comparison Benchmarks



SVM provided the highest accuracy by identifying subtle linguistic boundaries between factual reporting and hyperbolic misinformation.

Performance Metrics

- ✓ **Precision:** Ensuring that when we label a tweet as "Fake," it truly is. (Minimizing False Alarms).
- ✓ **Recall:** The ability to catch every single fake tweet in the stream. (Maximizing Sensitivity).
- ✓ **F1-Score:** The harmonic mean that balances both precision and recall for a robust evaluation.



Project Deliverables

- ✓ **Reproducible Jupyter Notebook:** A complete pipeline from data cleaning to final model evaluation.
- ✓ **Serialized Model:** A production-ready classifier stored as a pickle file for instant deployment.
- ✓ **Performance Report:** Visual documentation of confusion matrices and ROC curves for stakeholders.
- ✓ **Twitter API Integration:** Scripted logic to fetch and predict real-time tweet credibility.

Questions?

Thank you

